KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2011
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define digital signature. Write some properties of digital signature. (09)
   b) How RSA algorithm is used for digital signing? Explain. (08)
   c) Define the terms: cryptosystem, cryptanalysis. Discuss the benefits of asymmetric cryptosystem. (10)
   d) What is meant by homomorphic encryption? Explain with an example. (08)

2. a) Explain ElGamal encryption algorithm with an example. (13)
   b) What is meant by 'confusion' and 'diffusion'? Give an overview of DES algorithm. (09)
   c) Explain AES encryption algorithm. (09)
   d) Give an example of transposition cipher. (04)

3. a) In RSA algorithm, how $ed = 1$? [Where e and d have their usual meanings]. (07)
   b) What are the goals of security? Explain them briefly. (06)
   c) What are the parameters that can make a computer network vulnerable? Discuss them. (13)
   d) Discuss the requirements for database security. (09)

4. a) What is the purpose of encryption in a multilevel secure database management system? (08)
   b) Define sensitive data. What are the factors that can make data sensitive? Discuss them. (12)
   c) Explain the terms: i) denial-of-service (DoS) attack and ii) distributed denial-of-service (DDoS) attack. (09)
   d) What is cryptographic checksum? Explain with example. (06)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) "Security measures must balance between security and productivity"- Justify the statement. (08)
   b) Define risk analysis. Discuss the steps of risk analysis. (10)
   c) What is security planning? Write down the issues of security planning. (10)
   d) How does operating system offer protection at several levels? (07)

6. a) ~~What is~~ How virus gain control? Discuss with pictorial view. (10)
   b) What are the qualities of virus? What is virus signature and polymorphic virus? (12)
   c) "Computer crime is hard to define and hard to prosecute"- Justify. (13)

7. a) Discuss about exhaustive attack and one time password. (10)
   b) How buffer overflow affects code and data? Discuss elaborately. (15)
   c) What is Computer Virus? Differentiate between Virus and Worm. How Virus is attached? (10)

8. a) Write short notes on- (i) Risk exposure (ii) Risk leverage. (10)
   b) How fence register can be used to provide extra benefit in relocation method? What is the advantage of two pair base and bound register? (12)
   c) Discuss the tagged architecture with pictorial view. (13)

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